Comsats University Islamabad (Attock Campus)

Assignment no.03

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Course: Information Security Lab

Question no.01:

TASK 1 — Basic Blockchain Implementation

Introduction

Blockchain is a secure method of storing data in the form of blocks. Every block is connected to the previous block using cryptographic hashes. This makes blockchain very secure because changing one block affects all the blocks after it.

In this task, a simple blockchain was created in Python. Each block contains:

- Index number
- Timestamp
- Transaction data
- Previous block hash
- Current block hash

SHA256 hashing algorithm was used to generate hashes.

Importing Libraries:

```
import hashlib
import datetime
```

The hashlib library is imported to generate SHA256 hashes.
The datetime library is used to store the current date and time for each block.

Creating the Block Class:

```
class Block:
```

A class named Block is created to represent each block in the blockchain.

Every block stores important information such as transaction data and hashes.

Constructor of Block:

```
def __init__(self, index, timestamp, data, previous_hash):
```

This constructor initializes the values of a block.

- index stores the block number.

- timestamp stores the creation time.
- data stores transaction information.
- previous_hash stores the hash of the previous block.

After storing these values, the current block hash is calculated.

Calculating Hash:

self.hash = self.calculate_hash()

Each block generates its own hash value.

The hash acts like a digital fingerprint. Even a very small change in data creates a completely different hash.

SHA256 Hash Function:

return hashlib.sha256(block_string.encode()).hexdigest()

SHA256 converts the block data into a fixed-length secure hash.

This ensures:

- Data integrity
- Security
- Tamper detection

If someone changes transaction data, the hash also changes immediately.

Creating Blockchain Class:

class Blockchain:

This class manages the entire blockchain.

It is responsible for:

- Creating blocks
- Connecting blocks
- Validating the chain

Genesis Block:

create_genesis_block()

The first block in blockchain is called the Genesis Block.

It is special because:

- It has no previous block
- Its previous hash is set to 0

Every blockchain always starts with a genesis block.

Adding New Blocks:

add_block()

This function adds new blocks into the blockchain.

Before adding:

- The new block receives the hash of the previous block.
- Then its own hash is calculated.

This creates a secure chain connection between blocks.

Displaying Blockchain:

display_chain()

This function prints all block information on the screen.

For every block, the following details are displayed:

- Index
- Timestamp
- Data
- Previous hash
- Current hash

Blockchain Validation:

is_chain_valid()

This function checks whether the blockchain is secure or not.

It verifies:

1. Whether block hashes are correct

2. Whether previous hashes are properly linked

If someone modifies data in any block, the blockchain becomes invalid.

Demonstrating Blockchain Tampering:

In the program, one block transaction was changed from:

"Ahmed sends 2 BTC to Sara"

to:

"Ahmed sends 100 BTC to Sara"

After changing the data:

- The block hash changed
- Blockchain validation failed

This shows that blockchain can easily detect tampering.

Security Analysis of Blockchain:

Blockchain provides strong security because every block depends on the previous block hash.

Important security features include:

1. Immutability

Once data is added, it cannot be changed easily.

2. Data Integrity

SHA256 hashing protects data from modification.

3. Tamper Detection

Any small modification breaks the blockchain structure.

4. Transparency

Every block can be verified independently.

Because of these features, blockchain is used in:

- Cryptocurrency
- Banking
- Supply chain systems

- Digital transactions

BlockChain.py > Block > calculate_hash

```python
import hashlib
import datetime

# ----------------------------
# Block Class
# ----------------------------
class Block:

    def __init__(self, index, timestamp, data, previous_hash):

        self.index = index
        self.timestamp = timestamp
        self.data = data
        self.previous_hash = previous_hash

        # Calculate hash of current block
        self.hash = self.calculate_hash()

    # Function to calculate SHA256 hash
    def calculate_hash(self):

        block_string = (
            str(self.index) +
            str(self.timestamp) +
            str(self.data) +
            str(self.previous_hash)
        )

        return hashlib.sha256(block_string.encode()).hexdigest()


# ----------------------------
# Blockchain Class
# ----------------------------
class Blockchain:

    def __init__(self):

        self.chain = [self.create_genesis_block()]

    # Create first block
    def create_genesis_block(self):

        return Block(
            0,
            datetime.datetime.now(),
            "Genesis Block",
            "0"
        )

    # Get latest block
    def get_latest_block(self):

        return self.chain[-1]

    # Add new block
    def add_block(self, new_block):

        new_block.previous_hash = self.get_latest_block().hash

        new_block.hash = new_block.calculate_hash()
```

```python
        new_block.hash = new_block.calculate_hash()

        self.chain.append(new_block)

    # Display blockchain
    def display_chain(self):

        for block in self.chain:

            print("\n------------------------")
            print("Block Index:", block.index)
            print("Timestamp:", block.timestamp)
            print("Data:", block.data)
            print("Previous Hash:", block.previous_hash)
            print("Current Hash:", block.hash)

    # Verify blockchain integrity
    def is_chain_valid(self):

        for i in range(1, len(self.chain)):

            current_block = self.chain[i]
            previous_block = self.chain[i - 1]

            # Recalculate hash
            if current_block.hash != current_block.calculate_hash():

                return False

            # Compare previous hash
```

```python
            if current_block.previous_hash != previous_block.hash:

                return False

        return True


# ----------------------------
# Main Program
# ----------------------------

# Create blockchain
my_blockchain = Blockchain()

# Add blocks
my_blockchain.add_block(
    Block(1, datetime.datetime.now(), "Ali sends 5 BTC to Ahmed", "")
)

my_blockchain.add_block(
    Block(2, datetime.datetime.now(), "Ahmed sends 2 BTC to Sara", "")
)

my_blockchain.add_block(
    Block(3, datetime.datetime.now(), "Sara sends 1 BTC to Zoya", "")
)

my_blockchain.add_block(
    Block(4, datetime.datetime.now(), "Zoya sends 3 BTC to Hamza", "")
)
```

```python

# Display blockchain
print("\n========= ORIGINAL BLOCKCHAIN =========")
my_blockchain.display_chain()

# Validate blockchain
print("\nIs Blockchain Valid?", my_blockchain.is_chain_valid())

# ----------------------------------------
# Modify one block to show blockchain effect
# ----------------------------------------

print("\n\n========= MODIFYING BLOCK 2 =========")

my_blockchain.chain[2].data = "Ahmed sends 100 BTC to Sara"

# Recalculate modified block hash
my_blockchain.chain[2].hash = my_blockchain.chain[2].calculate_hash()

# Display blockchain again
my_blockchain.display_chain()

# Validate again
print("\nIs Blockchain Valid After Modification?",
      my_blockchain.is_chain_valid())
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\SHOAIB\Desktop\information security> & C:/Users/SHOAIB/AppData/Local
/Programs/Python/Python314/python.exe "c:/Users/SHOAIB/Desktop/information secur
ity/BlockChain.py"

========= ORIGINAL BLOCKCHAIN =========

------------------------
Block Index: 0
Timestamp: 2026-05-03 18:28:33.726227
Data: Genesis Block
Previous Hash: 0
Current Hash: 3d4925294e3d7e14b1a1cccca5a687df60a6cfb7989e3c5d0fb056d530d7fb8e

------------------------
Block Index: 1
Timestamp: 2026-05-03 18:28:33.726773
Data: Ali sends 5 BTC to Ahmed
Previous Hash: 3d4925294e3d7e14b1a1cccca5a687df60a6cfb7989e3c5d0fb056d530d7fb8e
Current Hash: 403598ce98a93e606880156be0d870f421515c7d1ac4cd4eee9a9f3c46f54257

------------------------
Block Index: 2
Timestamp: 2026-05-03 18:28:33.726810
Data: Ahmed sends 2 BTC to Sara
Previous Hash: 403598ce98a93e606880156be0d870f421515c7d1ac4cd4eee9a9f3c46f54257
Current Hash: 9ca33f7fab2fa733266f42b9e9a4a2569795a824bfda66d463065037eb3f3b0c

------------------------
Block Index: 3
Timestamp: 2026-05-03 18:28:33.726827
Data: Sara sends 1 BTC to Zoya
Previous Hash: 9ca33f7fab2fa733266f42b9e9a4a2569795a824bfda66d463065037eb3f3b0c
Current Hash: 19ffcdec0591f892f8bb9a6933b8b1b5d3f30ea7e9d37491aeb2365aa039c498
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

--------------------------
Block Index: 4
Timestamp: 2026-05-03 18:28:33.726840
Data: Zoya sends 3 BTC to Hamza
Previous Hash: 19ffcdec0591f892f8bb9a6933b8b1b5d3f30ea7e9d37491aeb2365aa039c498
Current Hash: 2f047aaa2075111f5d1ae0be9f0867027712d08ae23fab7763fda5343e6d422a

Is Blockchain Valid? True


========= MODIFYING BLOCK 2 =========

--------------------------
Block Index: 0
Timestamp: 2026-05-03 18:28:33.726227
Data: Genesis Block
Previous Hash: 0
Current Hash: 3d4925294e3d7e14b1a1cccca5a687df60a6cfb7989e3c5d0fb056d530d7fb8e

--------------------------
Block Index: 1
Timestamp: 2026-05-03 18:28:33.726773
Data: Ali sends 5 BTC to Ahmed
Previous Hash: 3d4925294e3d7e14b1a1cccca5a687df60a6cfb7989e3c5d0fb056d530d7fb8e
Current Hash: 403598ce98a93e606880156be0d870f421515c7d1ac4cd4eee9a9f3c46f54257

--------------------------
Block Index: 2
Timestamp: 2026-05-03 18:28:33.726810
Data: Ahmed sends 100 BTC to Sara
Previous Hash: 403598ce98a93e606880156be0d870f421515c7d1ac4cd4eee9a9f3c46f54257
Current Hash: b687eb357dd260f40c0e9a9f88efed5d2b3f93be8288c52737a58ddc47262fa7
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

--------------------------
Block Index: 3
Timestamp: 2026-05-03 18:28:33.726827
Data: Sara sends 1 BTC to Zoya
Previous Hash: 9ca33f7fab2fa733266f42b9e9a4a2569795a824bfda66d463065037eb3f3b0c
Current Hash: 19ffcdec0591f892f8bb9a6933b8b1b5d3f30ea7e9d37491aeb2365aa039c498

--------------------------
Block Index: 4
Timestamp: 2026-05-03 18:28:33.726840
Data: Zoya sends 3 BTC to Hamza
Previous Hash: 19ffcdec0591f892f8bb9a6933b8b1b5d3f30ea7e9d37491aeb2365aa039c498
Current Hash: 2f047aaa2075111f5d1ae0be9f0867027712d08ae23fab7763fda5343e6d422a

Is Blockchain Valid After Modification? False
PS C:\Users\SHOAIB\Desktop\information security>
```